Evaluation of the Western U.S. Power Grid Network and Improving Resilience Against Failures

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***Abstract*—The US power grid plays a critical role in supporting the daily life, economy, and national security, directly serving hundreds of millions of people. While the power gride is generally reliable nationwide, there are still possible opportunities for improvement at the local level, particularly in distribution reliability, rather than power generation, in some regions of the country [1] [2]. These vulnerabilities arise from random and targeted failures in the power grid, which can cause widespread outages. This work analyzes the U.S. power grid network, critical points, simulates failure scenarios, and explores potential strategies for improving its resilience. To better understand the relationships between power grid nodes, node2vec embeddings are applied through unsupervised learning [3], which provide insights into the network structure. For network improvement, cosine similarity is used to identify highly similar but unconnected nodes, which are then linked to enhance robustness. The original and modified networks are compared in performance under both random and targeted failure simulations. Results indicate that the modified network demonstrates greater resistance to fragmentation into disjoint subgraphs and has a larger connected component on average over many simulations. Although these findings suggest improved resilience, the cost of implementing these changes, in both time and money, is not evaluated here. Nonetheless, the proposed adjustments show a reduction in the theoretical network vulnerability.**

I. Introduction

THE United States power grid is one of the most critical infrastructures supporting modern society in North America. From powering homes and hospitals to enabling industry and communication systems, it is fundamental to national security and quality of life. While the U.S. power grid is generally considered highly reliable at the national scale, recent studies and real-world events suggest that some vulnerabilities still exist, especially in transmission lines and in certain regions that are prone to environmental disruptions [1], [2].

Large-scale blackouts, such as the North American blackout of 2003 and the Texas winter blackout of 2021, demonstrate how local failures can cascade through the power grid, resulting in widespread outages. These events have drawn increased attention to the vulnerability and fragility of these grids. As demand for power on the grid grows, the need for a more resilient network is important to consider.

Traditionally, network resilience has been evaluated through random and targeted node failure simulations, where nodes represent power substations or generation sites, and edges represent power lines. Random failures simulate events like weather or malfunctions, while targeted attacks model the removal of high-traffic nodes, which have a large effect on network integrity. While these approaches can highlight possible weaknesses, they do not provide a suggestion to how to improve the power grid.

In this project, we see how modern machine learning techniques, specifically node2vec embeddings, can be applied to enhance the resilience of the Western U.S. power grid network. Node2vec is a network embedding method that captures relationships between nodes by mapping them to a vector space, which allows for comparisons between nodes based on their relative position and role in the network. Using these embeddings, cosine similarity can be calculated for node pairings to identify potential new connections that could strengthen the network. The idea is to suggest new edges between nodes with similar neighbors or roles, potentially providing alternative power routes that reduce vulnerability.

To evaluate the effectiveness of these adjustments, we compare the original power grid network to a modified version that includes the proposed edges. Both networks are evaluated through simulations of random and targeted node failures. We measure resilience by the size of the largest connected component/subgraph and the number of disjoint subgraphs. The results show that the modified network is more robust, maintaining increased connectivity and resisting fragmentation more than the original network.

Although this study does not consider the cost of modifying the grid in real life, it provides an insight into identifying potential improvements in power grid networks. By applying machine learning for network analysis with more traditional failure simulation approaches, this work offers a new method for improving power grid resilience.

II. Related Work

Despite the US power grid's general reliability, major blackouts have highlighted the grid's vulnerability to both random and targeted failures. Thorough existing research has been performed to better understand these vulnerabilities and to propose methods for analyzing and improving grid resilience.

One essential related work is the final report on the August 14, 2003 blackout in the United States and Canada [1], which analyzed the causes and timeline of one of the most widespread blackouts in North American history. The report highlights a combination of human error, a lack of situational

awareness, and a lack of redundancy in transmission lines as primary causes of the outages. These findings highlight the importance of a robust network and the risks of relying on centralized nodes with few alternatives for rerouting.

Similarly, the 2021 Texas blackout, analyzed in ”Cascading Risks: Understanding the 2021 Winter Blackout in Texas” [2], illustrated how environmental attacks, such as extreme weather, can cause outages in a wide area even outside of the effected zones. The work stressed that vulnerabilities are primarily in transmission lines, but also in planning for these events beforehand. Both events make clear the room for improved resilience strategies that can adapt to unforeseen failures.

As an attempt to dampen these vulnerabilities, machine learning and graph embedding methods are a great way to propose new possible improvements. The node2vec algorithm [3] provides an effective approach to learn feature embeddings for nodes in a graph. By representing local and global connection information, node2vec embeddings allow for similarity comparisons between nodes. In this study, node2vec embeddings are used to identify pairs of similar but unconnected nodes. A new network that includes new edges between them is created as an attempt to improve network resilience.

Besides this, extensive research has been done to understand vulnerabilities in power grids. In ”Structural Vulnerability of the North American Power Grid” [4], they model the power grid as a network and analyze how it performs against random and targeted removals of nodes. They found that the grid is extremely robust against random failures; however, it is much more vulnerable to targeted attacks, particularly based on node degree or centrality. This observation influenced the methodology of this study, particularly in the use of targeted failures on the highest-degree nodes during failure simulations. With that said, random failures are also simulated for comparison.

Another related work, ”Network Robustness and Fragility: Percolation on Random Graphs” [5], provides methods for analyzing network fragmentation when node failures occur. The use of the size of the largest connected component as a resilience metric is applied during evaluation. These metrics measure the impact of node removals on network integrity and directly compare the resilience of the original and modified networks.

Finally, the paper ”Complex Systems Analysis of Series of Blackouts” [6] highlights how small, local failures can lead to cascading failures due to the interconnected grid and transmission line capacities. These ideas reinforce the importance of network redundancy and alternative connections in power grids. It also influenced the strategy in this study to add new edges based on embedding similarity.

Together, these works form the key ideas and background of this study. For instance, node2vec embeddings are used to find highly similar nodes with large relative centrality, and then new edges are created between them if they did not exist. This approach was inspired by the previously discussed works.

III. Methodology

A. US Power Grid Network

The power grid network used in this study [7] was obtained from the Network Repository and represents the real-life power grid of the Western United States. Nodes in the graph correspond to components such as generators, substations, transformers, and consumers. Meanwhile, edges represent the power lines connecting them. The graph is sparse, with an average degree of 2.67 and a maximum degree of 19, primarily due to the large number of consumer nodes, which typically have a single edge. Further network characteristics are shown in Table I.

For a realistic application, edges should rarely be added if they include consumer nodes, especially if both are consumers. Methods to avoid such connections are discussed in the *node2vec-based Edge Proposals* section. For a deeper understanding of the network’s structure and how each community is impacted by adjustments, the graph is also partitioned into communities using the Louvain algorithm, which is a modularity-based method for community detection. An overview of the graph with its 39 communities is shown in Figure 1.

TABLE I
US Power Grid Network Details

Total Nodes	4,941
Total Edges	6,594
Density	0.00054
Average Clustering Coefficient	0.08010
Average Degree	2.67
Maximum Degree	19

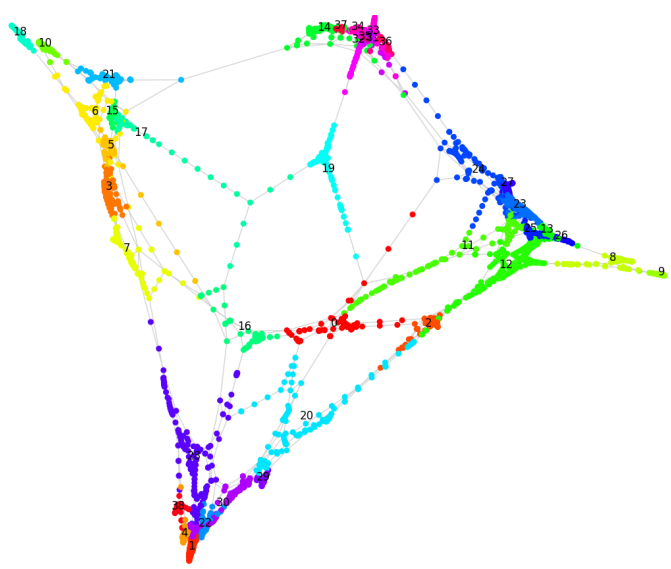

Fig. 1. Overview of US Power Grid Network, with each color set representing a different community found through the Louvain method for community detection.

B. node2vec-based Edge Proposals

To improve grid resilience to failures, this study proposes adding edges between highly similar nodes using node2vec

embeddings and cosine similarity. If two nodes are highly similar with many similar neighbors but unconnected, adding an edge may reduce load on the shared neighbors and improve network robustness against failures.

To prevent unnecessary connections between consumers, a betweenness centrality threshold is applied, which ensures edges are only added if both nodes exceed a given centrality. This prioritizes connections between more central nodes, enhancing redundancy for high-impact components rather than low-priority consumers which should not be assigned new edges.

C. Evaluation Methods

Network improvements are evaluated through repeated random and targeted failure simulations. As prior studies show, the U.S. power grid is resilient to random failures but vulnerable to targeted ones [4], so both are tested.

In each failure simulation type, nodes are removed along with their edges. Random failures are simulated in increments of one node from one up to 100 nodes, with 200 simulations per step. Targeted failures remove a random node with the highest degree, also with 200 simulations per step.

Additionally, network and community density and average clustering coefficients before and after modifications are measured to assess for improvements in redundancy and reliability.

IV. Implementation and Experiments

A. Implementation Details

The original power grid network used in this study is the *Western US Power Grid* dataset obtained from Network Repository [7]. The graph comprises 4,941 nodes and 6,594 edges, and each node represents a physical component of the grid such as generators, substations, transformers, and consumers. Network analysis was conducted using the `NetworkX` Python library, network visualization relied on `Matplotlib`, while `node2vec` embeddings were generated using the implementation specified in the original paper [3]. All experiments were executed on a local machine.

Before generating embeddings, the network was divided into communities using the Louvain algorithm to aid in visual interpretation and local analysis of different regions. The embeddings were generated using a walk length of 30, 200 walks per node, embedding dimensions of 64, and return parameters $p = 1$ and $q = 1$.

B. Edge Proposal and Filtering Process

To propose new edges, cosine similarity was calculated between all node pairs using their node2vec embeddings. Pairs that were already connected were excluded, and the most similar unconnected node pairs were considered for edge addition. The similarity threshold was set to a cosine similarity of greater than 0.9 to be considered, to filter out any node pairings that were not highly similar. To reduce the chance of connecting two consumer nodes (which would have limited real-world application), node pairs were filtered by a betweenness centrality threshold. Only pairs in which both nodes exceeded the 90th percentile of betweenness centrality were considered for new edges.

After filtering, a final set of 851 new edges was selected and added to the original graph to produce the modified power grid network. The thresholds also act as a thinning of the number of edges to be added, aiming for performance improvements while preserving the sparse structure of the original grid and not requiring an unrealistically large number of new edges.

C. Failure Simulation Setup

Two types of failure simulations were performed: random and targeted. For both, nodes were removed from the network along with their edges, and the size of the largest connected component and the number of disjoint subgraphs were recorded after each set of removals. This process was carried out on both the original and the modified network to compare resilience.

- **Random Failures:** For each removal count from 1 to 100, 200 random simulations were conducted. Nodes were randomly selected and removed, and the average size of the largest component and number of subgraphs were computed for each failure count. This resulted in a total of 20,000 random simulations per graph.
- **Targeted Failures:** For targeted failures, nodes with the highest degree were iteratively removed. Connections were broken randomly when multiple nodes shared the highest degree. As with random failures, 200 simulations were run for each node removal count, totaling 20,000 targeted simulations per graph.

D. Additional Metrics

To compare the structural differences of the two networks, the following network metrics were calculated for both:

- **Global Density:** Indicates how many edges exist compared to the maximum number possible for the entire network. An increase in suggests improved interconnectivity, resilience to failures, and redundancy across the network as a whole.
- **Global Average Clustering Coefficient:** Measures how well each node's neighbors are connected to each other, which correlates with failure resilience in the network. This is an average for all nodes in the entire network.
- **Local Density:** Indicates how many edges exist compared to the maximum number possible for each community.
- **Local Average Clustering Coefficient:** The same as the global average clustering coefficient, but only considers a single community. This metric is calculated for each community to analyze how each community improved in terms of connectivity.

These metrics were used to evaluate the redundancy and resilience to failures of the network on top of random and targeted failure simulations.

E. Visualization and Community Analysis

The final graphs and charts were visualized with color-coded communities that were found using Louvain community

detection. This allowed deeper insights into how each community changed, such as their densities and clustering coefficients, rather than just understanding the global network impacts of the modifications.

V. Results and Analysis

When including the new edges in the graph, the modified graph sees a 29.6% increase in density, and a 166.5% increase in clustering coefficient to the original graph. This is at the expense of adding 851 new nodes, or increasing the size of the Western US Power Grid by 12.9%. The increase in density and clustering coefficient highly suggest that the new network has increased interconnectivity and redundancy that outweighs the percentage increase of edges. While not definitive on their own, these combined with the later failure simulations show a clear advantage to the new network.

TABLE II
US Power Grid Network Details - Before & After Modifications

	Original	Proposed	% Increase
Total Nodes	4,941	4,941	0%
Total Edges	6,594	7,445	12.9%
Density	0.00054	0.00061	29.6%
Average Clustering Coefficient	0.08010	0.21346	166.5%
Average Degree	2.67	3.01	12.7%
Maximum Degree	19	23	21.1%

In Figures 2 and 3, community-level densities (and average density for all communities) are shown before and after modifications respectively. Between the charts, a clear increase in each community's density is displayed, with the graph exhibiting less skewed behavior.

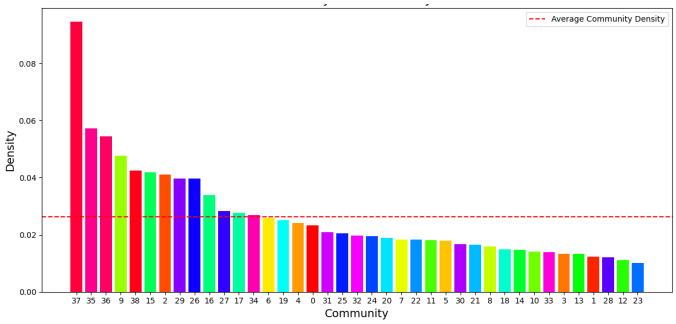

Fig. 2. Densities for communities of original graph. Average density for all communities is displayed with the dashed red line.

Similar to the densities, the clustering coefficients before and after also significantly increase, even more in fact. In Figures 4 and 5, the before and after clustering coefficients for each community is shown respectively. In this case, there is a clear reduction in the disparity of clustering coefficients between communities, indicating that communities with considerably low clustering coefficients were drastically improved, more so than communities with already high values. This suggests that interconnectivity was increased more in communities that needed it more compared to communities with already high connectivity.

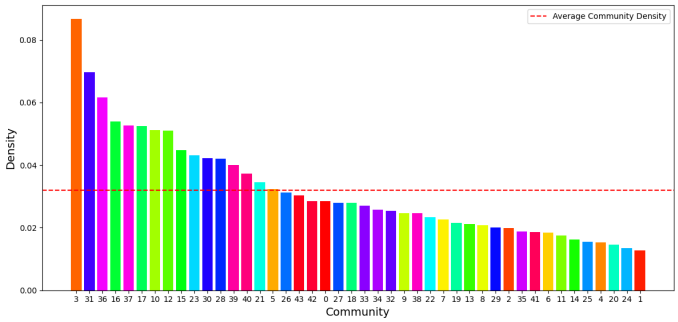

Fig. 3. Densities for communities of proposed graph with new edges. Average density for all communities is displayed with the dashed red line.

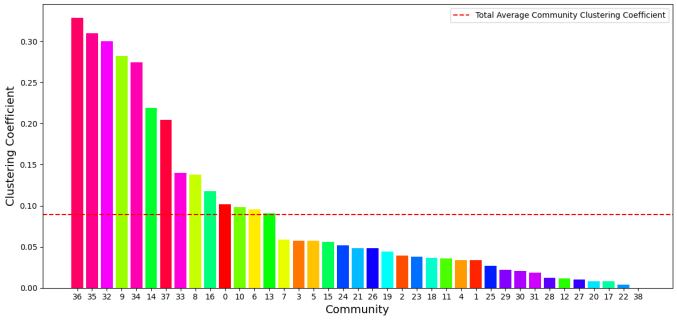

Fig. 4. Average clustering coefficients for communities of original graph. Average of the clustering coefficients for all communities is displayed with the dashed red line.

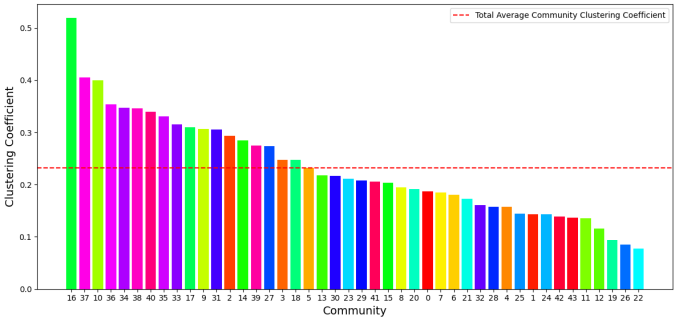

Fig. 5. Average clustering coefficients for communities of proposed graph with new edges. Average of the clustering coefficients for all communities is displayed with the dashed red line.

As for failure simulations, both random and targeted approaches were tested as previously described. The results for random simulations indicate improved connectivity through a higher average size of the largest connected component as more nodes are removed, as apparent in Figure 6. It is important to note that the rate that the size of the largest subgraph decreases is roughly 9.2% better for the new graph (found by comparing rate of change of the lines of best fit).

Similarly, the random simulations yielded an even more exaggerated improvement in the number of disjoint subgraphs formed as more nodes are removed. In Figure 7, it is clear that there is significantly less fragmentation for the new graph, indicating that communities are less susceptible to being disconnected from others, and that communities are more resilient to being fragmented. For instance, the rate that the

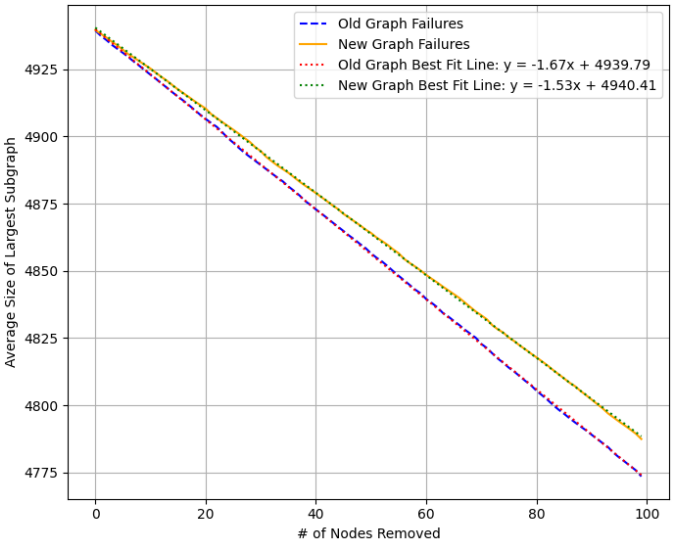

Fig. 6. Average sizes of largest connected components for random failure simulations. A set of simulations from 1-100 node removals are displayed.

number of disjoint graphs increases for the new network is 77.1% of the original graph, and a smaller rate is better for connectivity.

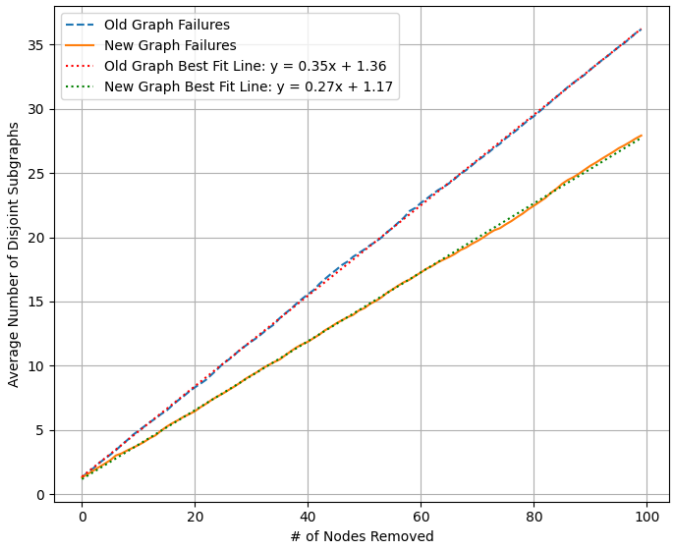

Fig. 7. Average number of disjoint subgraphs for random failure simulations. A set of simulations from 1-100 node removals are displayed.

Moving on to targeted simulations, these results indicate the new network experienced increased resilience against attacks. For the average size of the largest subgraph, the performance of the new graph is especially better after around 40 nodes removed. At around 50 nodes removed, the original graph quickly experienced a drop in the size of the largest subgraph, indicating that a large number of nodes (around 100), possibly an entire community, became completely disconnected from the main power grid. However, the new graph never experiences this massive fragmentation, at least not for targeted attacks on up to 100 nodes.

Additionally, the performance of the new network showed

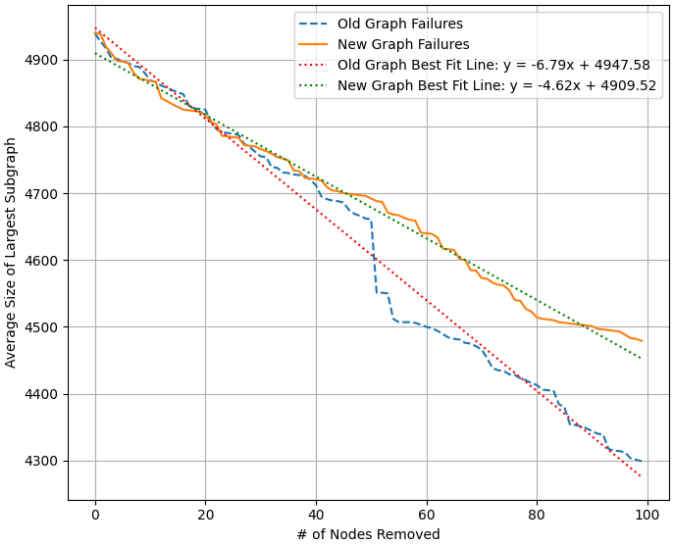

Fig. 8. Average sizes of largest connected components for targeted failure simulations. A set of simulations from 1-100 node removals are displayed.

a clear increase in resilience against targeted attacks in terms of fragmentation. The number of disjoint subgraphs increases much more rapidly for the original network, around 68.7% faster in fact, compared to modified graph (based on rate of change of the line of best fit). These results directly show that less sets of nodes become completely disconnected from others with the proposed network, highly suggesting that less groups of consumers become disconnected from the larger power grid which risks cascading failures.

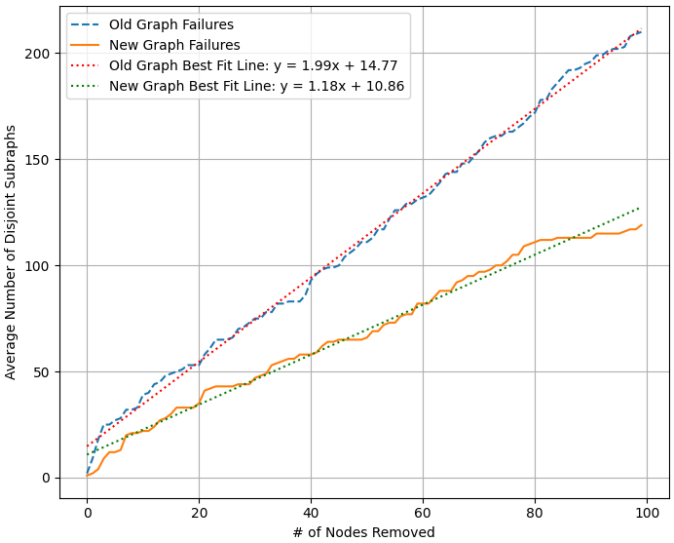

Fig. 9. Average number of disjoint subgraphs for targeted failure simulations. A set of simulations from 1-100 node removals are displayed.

VI. Conclusion

This study evaluated the resilience of the Western U.S. power grid by analyzing its communities and proposing possible edge additions based on node2vec similarity and betweenness centrality. The modified network, which added 12.9%

more edges, demonstrated clear improvements. Specifically, it achieved a 29.6% increase in density and a 166.5% increase in clustering coefficient, indicating enhanced connectivity and redundancy.

Failure simulation results further support the proposed adjustments. The modified network consistently maintained a larger connected component/subgraph and exhibited significantly less fragmentation during both random and targeted failures. Notably, in targeted simulations, the original graph showed signs of major disconnections in the grid after 50 node removals, which suggests the loss of entire communities, while the modified graph did not see this drop. Fragmentation, which is measured by the number of disjoint subgraphs, also increased 31.3% slower in the new network against targeted failures.

These outcomes suggest that the proposed changes significantly improve the grid's ability to withstand failures, both random and targeted. While this study does not consider the practical costs of implementing new transmission lines, the results strongly indicate that enhancing connections, especially between high-centrality nodes, can yield a more robust and resilient power grid.

References

[1] U.S.-Canada Power System Outage Task Force, "Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations," Apr. 2004.

[2] Joshua W. Busby, et al., "Cascading risks: Understanding the 2021 winter blackout in Texas," Energy Research & Social Science, July 2021.

[3] A. Grover, J. Leskovec, "node2vec: Scalable Feature Learning for Networks," arxiv, July 2016.

[4] R. Albert, I. Albert, and G. L. Nakarado, "Structural vulnerability of the North American power grid," Physical Review E, Jan. 2004.

[5] D. S. Callaway, M. E. J. Newman, S. H. Strogatz, and D. J. Watts, "Network robustness and fragility: Percolation on random graphs," Physical Review Letter, Dec. 2000.

[6] I. Dobson, B. A. Carreras, V. E. Lynch, D. E. Newman, "Complex systems analysis of series of blackouts: Cascading failure, critical points, and self-organization," ResearchGate, July 2007.

[7] D. Watts, S. Strogatz, "Collective dynamics of small-world networks", Nature, 2015.